

A MULTIFRACTAL AND REGIME-BASED FORECASTING FRAMEWORK FOR BRENT CRUDE AND BDTI RETURNS

A PROJECT REPORT

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TABLE OF CONTENTS

ABSTRACT

1. INTRODUCTION	2
1.1 Motivation	2
1.2 Objectives.....	3
1.3 Problem Statement	3
2. BACKGROUND AND LITERATURE REVIEW	5
2.1 Multifractal Analysis in Finance.....	5
2.2 BDTI as a Predictive Indicator.....	5
2.3 Regime Detection in Commodity Markets	6
3. DATASET AND PREPROCESSING.....	8
3.1 Data Sources.....	8
3.2 Log Return Computation.....	9
3.3 Descriptive Statistics.....	9
4. METHODOLOGY	11
4.1 Multifractal Detrended Fluctuation Analysis (MF-DFA).....	12
4.2 Structural Break Detection and Regime Classification.....	14
4.3 Feature Engineering and Data Preparation	15
4.4 Machine Learning Models	15
4.4.1 Ensemble Models.....	15
4.4.2 Baseline Models.....	16
4.5 Evaluation Metrics	16
5. RESULTS AND DISCUSSION.....	18
5.1 Overall Model Performance (Full Test Set).....	18
5.2 Feature Importance Analysis.....	19
5.3 Regime-wise Performance Analysis	20
5.4 Volatility-Stratified Analysis	22
5.5 Cross-Correlation Analysis	23

5.6 Discussion	23
6. CONCLUSION AND FUTURE WORK.....	26
6.1 Conclusion.....	26
6.2 Future Work	26
7.REFERENCES	28

LIST OF FIGURES

Fig 3.1 Market regimes identified using segmentation.....	8
Table 3.3 Descriptive Statistics of the Dataset.....	9
Fig 4 Daily time series of Brent crude oil and BDTI.....	11
Fig 4.1 MF-DFA fluctuation functions.....	12
Fig 4.1.1 Generalized Hurst Exponents: BDTI vs Brent.....	13
Table 4.2 Overall Performance - Random Forest (Full Test Set)	14
Table 4.3 Feature Configurations.....	15
Table 5.1 Overall Model Performance.....	18
Table 5.1.1 Model Comparison - Full Feature Set (Model C, Full Test Set)	18
Fig 5.1 Model Performance across Feature Configurations.....	19
Table 5.2 Feature Importance-Random Forest (Model C)	19
Fig 5.2 Feature Importance Comparison.....	20
Table 5.3 Regime-wise MSE -Model B vs Model C (All Models)	21
Table 5.3.1 Average Regime-wise MSE - Model B vs Model C (Random Forest)	21
Fig 5.3 MSE Improvement % by Regime and Model.....	22
Table 5.4 Performance by Volatility Regime -Random Forest.....	22
Table 5.5 Pearson Correlation - Brent Return vs BDTI Return by Regime.....	23

ABSTRACT

Financial markets are inherently complex, non-linear, and non-stationary systems, making reliable forecasting challenging. This study develops a regime-based multifractal forecasting framework to predict returns on Brent crude oil by integrating classical lag-based predictors, shipping market dynamics captured by the Baltic Dirty Tanker Index (BDTI), and multifractal complexity measures derived using Multifractal Detrended Fluctuation Analysis (MF-DFA). The multifractal spectrum width ($\Delta\alpha$), computed using a 250-day rolling window, is used as a proxy for time-varying market complexity. Binary segmentation is applied to detect structural breaks and identify six distinct market regimes. The framework is implemented using machine learning models, including Random Forest, XGBoost, LightGBM, and CatBoost, evaluated on progressively enriched feature sets. Empirical results show that multifractal features significantly improve forecasting performance, particularly during high-volatility and crisis periods, reducing the Mean Squared Error (MSE) of the CatBoost model by up to 64%, while improvements are marginal during stable regimes. Model performance is evaluated using MSE, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE), as the coefficient of determination (R^2) is less informative for financial return prediction due to the low signal-to-noise ratio. Among the models, CatBoost achieves the best performance with the lowest MSE of 0.000334, outperforming classical econometric models such as ARIMA and GARCH. Overall, the results highlight the regime-dependent nature of financial markets and show that multifractal features act as conditional predictors, particularly during periods of market stress.

CHAPTER 1

INTRODUCTION

1. INTRODUCTION

In particular, financial time series data, especially related to commodity markets such as crude oil prices, show very nonstationary behavior with features of volatility clustering, fat tail distribution, nonlinearity, and structural changes. Classic models, such as ARIMA and GARCH, assume linear relationships and stationarity in time series data, which is often not met in practical applications. Therefore, classic models have difficulty capturing the dynamics of financial markets, especially during crisis periods.

The Brent crude oil market depends on various global events, including geopolitical crises, supply and demand imbalance, and logistical issues. One of the key exogenous indicators for the Brent oil market is the Baltic Dirty Tanker Index (BDTI), which is calculated using the price of dry bulk shipping freights for transporting crude oil. Linear relations among such variables may be insufficient, indicating some kind of hidden nonlinear structure.

Therefore, this research paper presents a hybrid framework that combines multifractal complexity and machine learning models. Specifically, the proposed approach involves integrating multifractal complexity measures, obtained through multifractal detrended fluctuation analysis (MF-DFA), into the ML models' training process to account for hidden nonlinearity. The dataset used in this study covers the period from March 2013 to April 2026, comprising about 3,250 observations of daily oil prices within six market regimes.

1.1 Motivation

Firstly, the key driver behind this research is the fact that financial returns can be considered a near random walk process, which makes forecasting highly challenging under a standard framework. However, underlying these random-like movements are intricate scaling patterns and interdependencies that can be harnessed through multifractal theory. Such patterns tend to become even more visible during volatile times when the markets are stressed. In addition, information from outside measures like shipping indexes may also provide an insight that is ignored in a typical model.

1.2 Objectives

- Design an advanced forecasting method combining the properties of multifractality and machine learning to forecast returns of Brent crude oil.
- Determine the marginal value of the multifractal spectrum width $\Delta\alpha$ to predictability for three increasingly complex combinations of features.
- Study the influence of different market regimes recognized through binary segmentation on forecasting accuracy.
- Assess the forecasting capabilities of four ensemble gradient-boosting algorithms (Random Forest, XGBoost, LightGBM, and CatBoost), compared to other benchmarks.

1.3 Problem Statement

Even with the advent of financial modelling, the prediction of returns still poses a problem because of noise, nonlinearities, and changes in regimes. In this research, it will be analysed whether multifractality features, especially the range of $\Delta\alpha$, can improve forecasting capabilities once integrated with machine learning algorithms, and whether such improvement holds across varying regimes. The core assumption is that $\Delta\alpha$ serves as a conditional enhancer whose effectiveness is most significant during crises and uncertainty regimes.

CHAPTER 2

BACKGROUND AND LITERATURE REVIEW

2. BACKGROUND AND LITERATURE REVIEW

It is well-known that financial markets are complex systems with multifractal dynamics wherein the scaling nature varies for various moments. The following subsections elaborate on the underlying theory for multifractals, the forecasting ability of freight indexes, and recent developments in machine learning applications for finance.

2.1 Multifractal Analysis in Finance

Multifractal Detrended Fluctuation Analysis (MF-DFA), developed by Kantelhardt et al. (2002), is widely used to perform multifractal analysis of non-stationary time series in finance. The generalised Hurst exponent $H(q)$ describes long memory properties of a time series. In case of $H(q)$ independence from q , a time series is considered monofractal; otherwise, it is multifractal.

Multifractal spectrum width, $\Delta\alpha = \alpha_{\max} - \alpha_{\min}$, describes heterogeneity of the time series; larger $\Delta\alpha$ value means higher complexity and turbulence in the market conditions. According to Chen et al. (2025), considering multifractal properties of a time series allows predicting Brent oil prices more accurately using machine learning; improvements in R^2 range between 55-182%, while MSE values decrease by 60-92% on four crisis cases. The proposed methodology in the paper of Chen et al. (2025) uses stacking regressions based on XGBoost, LightGBM, and CatBoost with a Ridge meta-learner.

2.2 BDTI as a Predictive Indicator

Baltic Dirty Tanker Index (BDTI) represents a daily index of freight rates for the carriage of crude oil. Classical economic thinking suggests that oil prices are responsible for influencing freight rates; however, recent literature has put forward contrary arguments. According to Siddiqui and Basu (2020), oil prices influence freight rates in medium-long run, while Shi et al. (2022) found that shocks in oil price unrelated to supply significantly impact freight rate behaviour. Chen et al.

(2025) have empirically confirmed that BDTI returns act as leading indicators of Brent prices, especially if supplemented with multifractal properties.

This research builds on this prior work by (i) extending the data set until April 2026, (ii) forecasting log-returns instead of prices, (iii) using rolling MF-DFA window in order to account for dynamic $\Delta\alpha$, and (iv) explicitly considering regimes through binarizing return, volatility, and $\Delta\alpha$ indicators.

2.3 Regime Detection in Commodity Markets

Many studies have shown evidence of structural changes and market regimes in commodity markets. There are several fast algorithms based on the Binseg procedure and PELT for detecting changes in financial data series. In the current paper, we use Binseg algorithm for detecting changes in the following signals: raw returns, 30-day volatility, and $\Delta\alpha$. This multivariate approach can minimize the problem of finding a spurious regime boundary due to one signal.

CHAPTER 3

DATASET AND PREPROCESSING

3. DATASET AND PREPROCESSING

3.1 Data Sources

The dataset consists of daily Brent crude oil prices and Baltic Dirty Tanker Index (BDTI) values, covering the period from 14 March 2013 to April 2026, yielding a total of 3,250 raw observations.

The data sources are:

- BDTI (2013–2023): Sourced from Investing.com (composite index).
- BDTI (2023–2026): Sourced from the official Baltic Exchange via data request.
- Brent crude oil prices: FRED database (series code: DCOILBRENTU).

The combined dataset covers multiple full economic cycles including the 2014–16 oil price crash, the 2018–19 pre-COVID instability period, the 2020 COVID-19 crisis, the post-pandemic recovery, and the 2022 Russia–Ukraine geopolitical shock (the timeseries is shown in Fig 3.1)



Fig 3.1 Time series of BDTI and Brent Crude

3.2 Log Return Computation

Price series are transformed into logarithmic returns to ensure approximate stationarity:

$$r_t = \ln(P_t / P_{t-1})$$

where P_t is the closing price on day t . Log returns are preferred over simple returns in multifractal literature because they are additive over time and their distribution is closer to symmetry.

3.3 Descriptive Statistics

Table 3.3 presents descriptive statistics for the final processed dataset after return computation and cleaning.

Statistic	BDTI Return	Brent Return	$\Delta\alpha$
Count	2988	2988	2988
Mean	6.4×10^{-5}	-1.2×10^{-5}	0.691
Std. Dev.	0.024	0.028	0.279
Min	-0.38	-0.28	-0.340
Max	0.24	0.20	1.401

Table 3.3: Descriptive Statistics of the Dataset

CHAPTER 4

METHODOLOGY

4. METHODOLOGY

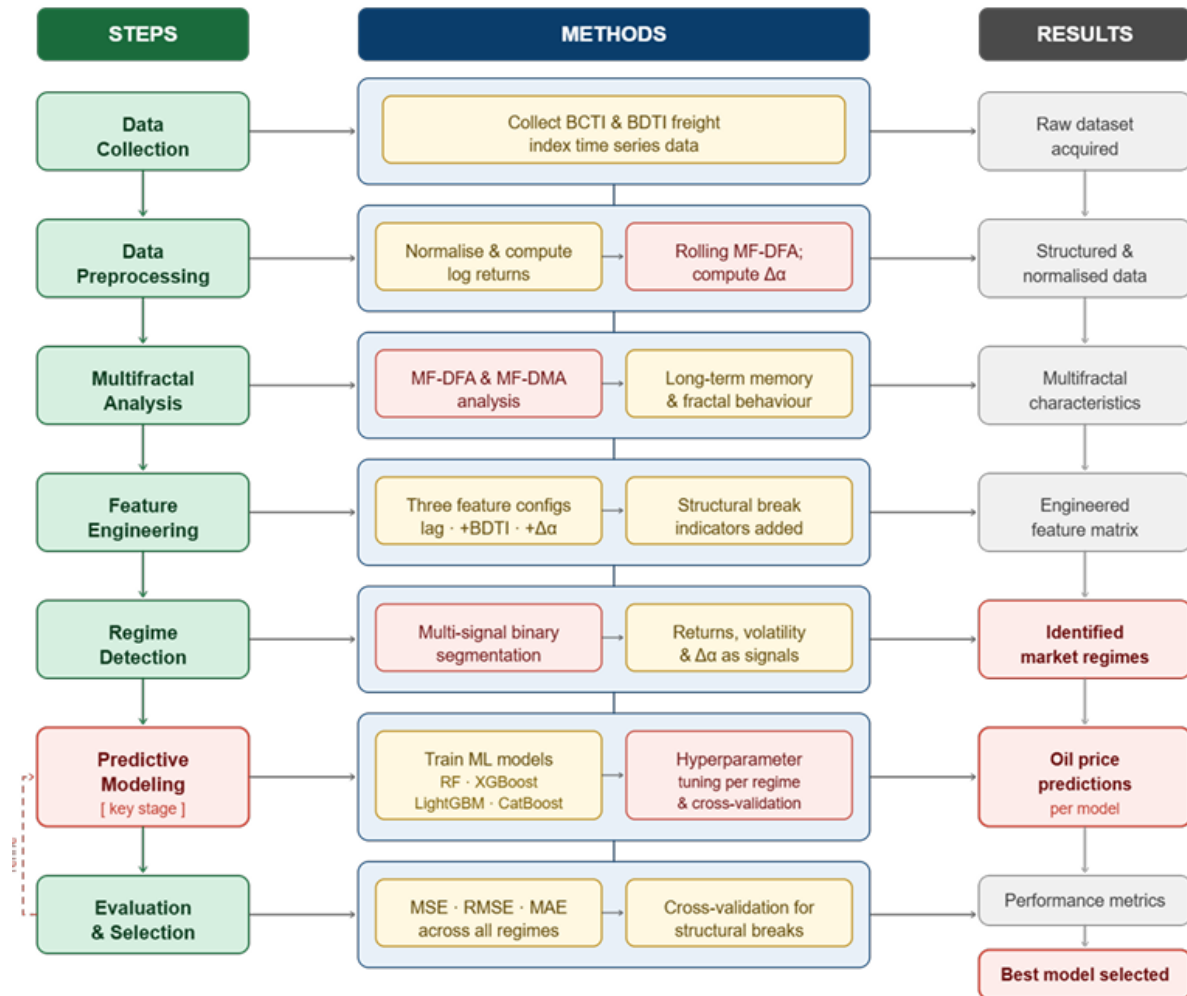


Fig 4 Methodology Framework

This figure 4 illustrates the end-to-end workflow of the proposed framework, outlining the sequential steps from data collection and preprocessing through multifractal analysis, feature engineering, regime detection, and machine learning-based predictive modelling, culminating in performance evaluation and optimal model selection.

The proposed framework is structured in four stages: (i) multifractal feature extraction using MF-DFA, (ii) regime identification using binary segmentation, (iii) feature engineering and model training, and (iv) regime-stratified evaluation. A rolling window approach is adopted throughout to ensure temporal integrity and avoid look-ahead bias.

4.1 Multifractal Detrended Fluctuation Analysis (MF-DFA)

To analyze the scaling properties of BDTI return time series, we use MF-DFA proposed by Kantelhardt et al. (2002). This method consists of several stages described below:

Step 1: Calculation of profile

A return series $x(i)$ of length N is transformed to the cumulative profile $Y(i)$:

$$Y(i) = \sum_{k=1}^i (x(k) - \bar{x}), i = 1, 2, \dots, N$$

where $\bar{x}$ is the sample mean. Thus, we transform the initial time series to the random walk series which allows detecting scaling.

Step 2: Segmentation and Detrending

Profile $Y(i)$ is segmented to equal pieces of size s . To utilize all data without losing them at borders, we repeat segmentation from both sides obtaining $2N_s$ segments. In each segment v , the local polynomial detrend is estimated and subtracted. The polynomial is linear, i.e. $m = 1$, and found using the least squares method. Local variances are calculated as follows:

$$F^2(v, s) = (1/s) \sum_{i=1}^s (Y((v-1)s + i) - P_v(i))^2$$

Step 3: q-order Fluctuation Function

All $2N_s$ pieces are averaged to receive q-order fluctuation function:

$$F_q(s) = \left\{ \frac{1}{(2N_s)} \sum_{v=1}^{(2N_s)} [F^2(v, s)]^{(q/2)} \right\}^{(1/q)}$$

$$F_0(s) = \exp \left\{ \frac{1}{(4N_s)} \sum_{v=1}^{(2N_s)} \ln [F^2(v, s)] \right\}$$

Parameter q regulates the weight put on large fluctuations (when $q > 0$) or small fluctuations (when $q < 0$). Here q varies in the set of $\{-5, -3, -1, 1, 3, 5\}$. Figure 4.1.1 presents the MF-DFA fluctuation functions, demonstrating scaling behavior across different q values.

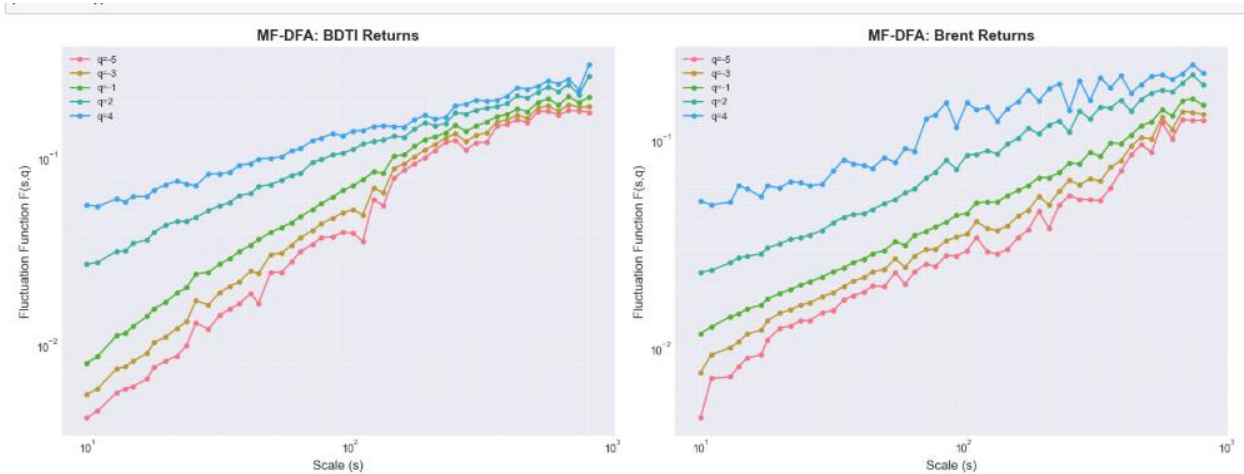


Fig 4.1.1 MF-DFA fluctuation functions

Step 4: Generalized Hurst Exponent

If the series exhibits long-range power-law correlations, $F_q(s)$ scales as:

$$F_{q(s)} \sim s^{H(q)}$$

The generalized Hurst exponent $H(q)$ is estimated as the slope in a log-log regression of $F_q(s)$ on s . For a stationary series, $H(2)$ reduces to the classical Hurst exponent H . For a multifractal series, $H(q)$ is strictly decreasing in q . Figure 4.1.2 compares the generalized Hurst exponents of Brent and BDTI, indicating multifractal characteristics in both series.

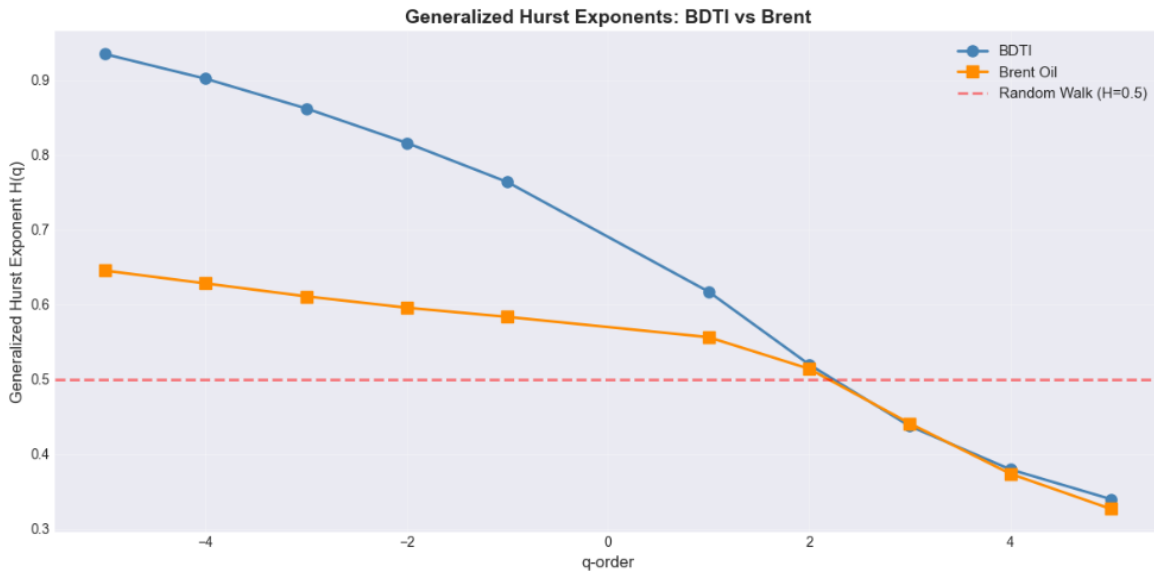


Fig 4.1.2 Generalized Hurst Exponents: BDTI vs Brent

Step 5: The width of the multifractal spectrum:

The multifractal spectrum $f(\alpha)$ can be derived using the Legendre transform below:

$$\tau(q) = qH(q) - 1$$

$$\alpha = d(\tau(q))/dq$$

$$f(\alpha) = q\alpha - \tau(q)$$

The width of the multifractal spectrum is:

larger the value of $\Delta\alpha$, the more heterogeneous the scaling behaviour is, which implies that the market condition is more turbulent and complicated.

Step 6: Implementing the rolling window:

For the current case, the rolling window is equal to $W = 250$ trading days (about one trading year).

This calculation is done for each 5 data points to balance computational effectiveness and temporal resolution.

$$\Delta\alpha_t = MF - DFA(\{r_{t-W+1}, \dots, r_t\})$$

The generated $\Delta\alpha$ sequence (mean = 0.691, std = 0.279, range = [-0.340, 1.401]) is lagged by one time period as an input feature to the model.

4.2 Structural Break Detection and Regime Classification

Market regimes are identified through a multi-signal binary segmentation (Binseg) approach applied separately to three signals: (i) raw Brent log-returns, (ii) 30-day rolling volatility σ_t , and (iii) rolling $\Delta\alpha$. For each signal, the algorithm minimises the total L2 cost across 5 break points:

$$C(\text{signal}) = \sum_{k=1}^{K+1} \sum_{t \in R_k} (\text{signal}_t - \mu_k)^2$$

where R_k is the k -th segment and μ_k its mean. Break dates from the three signals are then reconciled by choosing consensus dates with economic interpretability. The final six regime boundaries are:

Regime	Period	Description	n (obs)	Avg Volatility	Mean $\Delta\alpha$
1	Mar 2013 – Oct 2014	Pre-Oil Crash Stability	~100	0.010	0.688
2	Oct 2014 – Nov 2018	Oil Crash & Adjustment	1,010	0.023	0.753
3	Nov 2018 – Jan 2020	Pre-COVID Instability	295	0.022	0.656
4	Jan 2020 – Aug 2021	COVID-19 Crisis (High Volatility)	400	0.059	0.592
5	Aug 2021 –	Recovery & Ukraine	265	0.029	0.379

Table 4.2: Overall Performance — Random Forest (Full Test Set)

An ANOVA test confirms that $\Delta\alpha$ differs significantly across regimes (F-statistic = 121.11, p-value $\approx 5.86 \times 10^{-117}$), rejecting the null hypothesis of equal multifractal structure. This validates that the identified regimes represent genuinely distinct market states.

4.3 Feature Engineering and Data Preparation

The prediction target is the next-day Brent log-return: $r_t = \ln (P_t / P_{(t-1)})$

. Three feature configurations are constructed:

Model	Features	Description	Purpose
A	Brent Lag ₁	Lagged Brent return only	Baseline AR(1) proxy
B	Brent Lag ₁ , BDTI Lag ₁	Adds shipping market signal	Cross-market information
C	Brent Lag ₁ , BDTI Lag ₁ , $\Delta\alpha$ Lag ₁	Adds multifractal complexity	Full nonlinear model

Table 4.3: Feature Configurations

All variables are lagged by one period to avoid look-ahead bias. A 30-day volatility σ_t and the slope of the Brent trend using linear regression (250 days) are calculated as auxiliary variables. The data is divided into an 80%/20% training/testing sample based on chronological order (Training: ~2,390 observations; Test: ~598 observations).

4.4 Machine Learning Models

4.4.1 Ensemble Models

Four ensemble-based models are employed:

Random Forest: Consists of T trees, each tree being grown on a bootstrap sample of the data. Average prediction is taken across all trees: $y_{hat} = (1/T) \sum_{t=1}^T f_t(x)$ Variance reduction and overfitting prevention achieved by random subspace choice at each split. In our study, $T = 300$.

XGBoost: Regularized gradient boosting model minimizes $L(\varphi) = \sum_i l(\hat{y}_i, y_i) + \sum_k \Omega(f_k)$, where $\Omega(f) = \gamma T + (1/2)\lambda ||\omega||^2$. Efficient split calculation based on second-order Taylor series expansion. Model configuration: 300 estimators, learning rate 0.05, maximum tree depth 4.

LightGBM: Histogram-based GBDT using GOSS and EFB techniques for reducing computational complexity from $O(\text{data} \times \text{features})$ to $O(\text{data} \times \text{bundles})$. Model configuration: 300 estimators.

CatBoost: Ordered boosting implementation together with ordered TS approach for addressing the problem of prediction shift. Native handling of categorical variables. Model configuration: 300 iterations, random seed 42.

4.4.2 Baseline Models

Linear Regression is used as a simple baseline to quantify the linear predictability of the feature sets. ARIMA and GARCH benchmarks are included for comparison against traditional time series approaches.

4.5 Evaluation Metrics

Model performance is evaluated using three metrics:

$$MSE = \left(\frac{1}{n}\right) \sum_{t=1}^n (y_t - y_{hat_t})^2$$

$$RMSE = \sqrt{MSE}$$

$$MAE = \left(\frac{1}{n}\right) \sum_{t=1}^n |y_t - y_{hat_t}|$$

R^2 is not used for evaluation since return series in finance are roughly unpredictable noise. R^2 measures a comparison of a model to an unconditional mean of the process; for random walk-like return series, a correctly specified model will produce a negative or close to zero R squared measure. MSE and MAE directly compare error size and are widely used in the financial return forecasting literature.

Improvement in MSE by introducing $\Delta\alpha$ is measured as follows:

$$\text{Improvement (\%)} = (MSE_B - MSE_C) / MSE_B * 100$$

A positive number means that the introduction of $\Delta\alpha$ decreases error; a negative one means otherwise.

CHAPTER 5

RESULTS AND DISCUSSION

5. RESULTS AND DISCUSSION

5.1 Overall Model Performance (Full Test Set)

Table 5.1 presents the overall performance on the full test set for Random Forest under the three feature configurations, demonstrating the incremental contribution of BDTI and $\Delta\alpha$.

Model	Features	MSE	RMSE	MAE
Model A	Brent Lag ₁	0.000861	0.02935	0.02063
Model B	Brent Lag ₁ + BDTI Lag ₁	0.000696	0.02638	0.01828
Model C	Brent Lag ₁ + BDTI Lag ₁ + $\Delta\alpha$ Lag ₁	0.000632	0.02515	0.01748

Table 5.1: Overall Performance — Random Forest (Full Test Set)

The inclusion of BDTI in the base model improves MSE by 19.2% (Model A to Model B), whereas the inclusion of $\Delta\alpha$ improves MSE by 9.2% (Model B to Model C). Table 5.1.1 presents a comparison of all the four ML models with the full feature set.

Model	MSE	RMSE	MAE
CatBoost	0.000334	0.01828	0.01469
Random Forest	0.000364	0.01908	0.01533
LightGBM	0.000431	0.02077	0.01635
XGBoost	0.000478	0.02187	0.01737
Linear Regression (B)	0.000505	0.02247	0.01593
ARIMA	0.000551	0.02347	0.01712
GARCH	0.000551	0.02347	0.01712

Table 5.1.1: Model Comparison — Full Feature Set (Model C, Full Test Set)

CatBoost provides the highest accuracy overall, possibly because of its ordered boosting approach, which avoids any prediction shift – an especially important feature considering the non-stationarity of the data generation process in finance. ML methods clearly outperform traditional ARIMA and GARCH methods, thereby establishing their supremacy over the linear approach. Figure 5.1 visualizes the performance of models across different feature configurations, showing consistent improvement with feature enrichment.

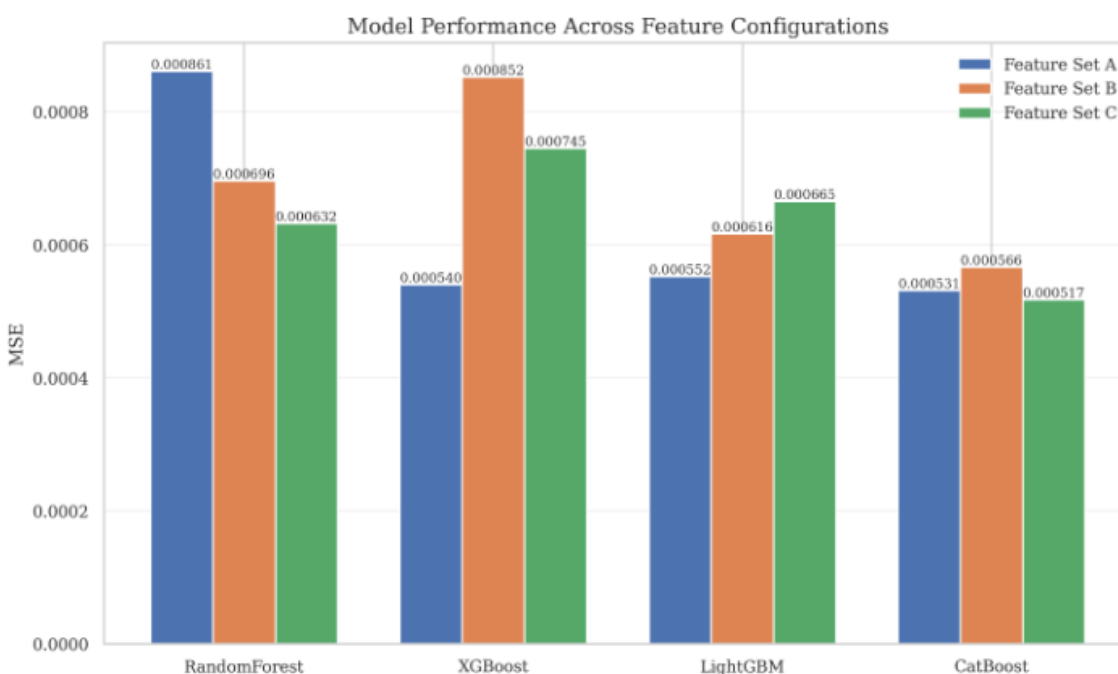


Fig 5.1 Model Performance across Feature Configurations

5.2 Feature Importance Analysis

Feature importance scores from the Random Forest model trained on the full feature set (Model C) are presented in Table 6. The scores represent the mean decrease in impurity (MDI) across all trees, normalized to sum to 1.

Feature	MDI Importance
Brent Lag ₁	0.516
BDTI Lag ₁	0.274
$\Delta\alpha$ Lag ₁	0.210

Table 5.2: Feature Importance — Random Forest (Model C)

The contribution of the lagged Brent return is 51.6%, which captures the momentum effect of oil return. The contribution of BDTI lag is 27.4%, representing cross-market shipping signals. $\Delta\alpha$ has a contribution rate of 21.0%, supporting that multifractal complexity contains useful forecasting information.

On the other hand, we observe that the result of XGBoost on full feature set shows a different hierarchy of feature importance as follows: BDTI Lag₁ (40.3%), $\Delta\alpha$ Lag₁ (36.4%), and Brent Lag₁ (23.2%). It shows that XGBoost exploits more nonlinear effects. (Shown in fig 5.2)

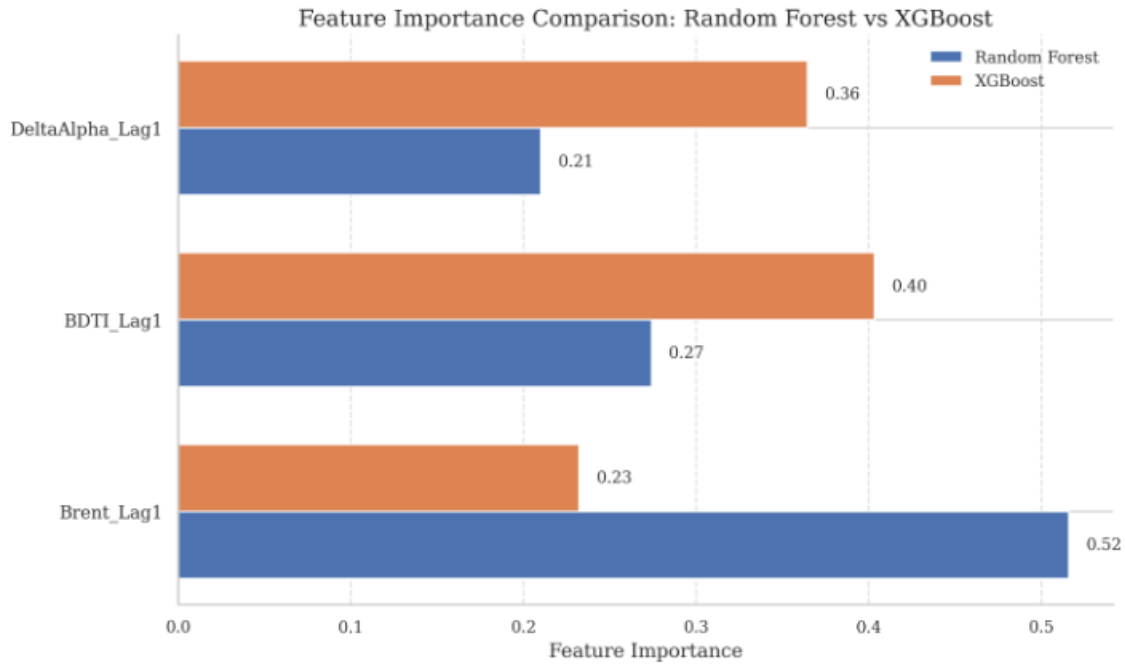


Fig 5.2 Feature Importance Comparison

5.3 Regime-wise Performance Analysis

The central contribution of this study is the regime-stratified evaluation. Table 7 presents the complete MSE results for both Model B and Model C across all four machine learning models and all regimes, along with the percentage improvement attributable to $\Delta\alpha$.

R#	Description	RF (C vs B%)	XGB (C vs B%)	LGB (C vs B%)	CAT (C vs B%)	Best Model (MSE↓%)
2	Oil Crash	+2.27	−13.21	−19.76	−12.21	RF +2.27%
3	Pre-COVID	+1.22	−14.66	−28.75	−1.50	RF +1.22%
4	COVID Crisis	+30.50	+64.09	−8.02	+12.54	XGB +64.09%
5	Recovery	−5.35	−17.11	−17.55	+6.30	CAT +6.30%
6	Post-War	+6.97	+3.09	+9.62	+7.06	LGB +9.62%

Table 5.3: Regime-wise MSE — Model B vs Model C (All Models)

Notably, Regime 4 (COVID Shock) yields an improvement of 64.09% in MSE for XGBoost due to $\Delta\alpha$. The reason behind such a large increase in performance stems from the fact that the COVID shock from January–August 2021 made the market highly volatile, thereby increasing its multifractality; hence, a larger discriminative power was achieved by $\Delta\alpha$. Random Forest provides an increase of 30.50%, whereas CatBoost improves by 12.54%. On the other hand, LightGBM displays a negative improvement in Regime 4.

Regimes 2 and 3 display minimal or negative improvements, which may be attributed to the fact that during stable-to-moderate periods, market dynamics can be approximated as a random walk. Hence, multifractal information becomes less meaningful. Regime 5 (Recovery/Ukraine Shock) also displays predominantly negative improvements in the MSE due to $\Delta\alpha$; this can be explained by the fact that $\Delta\alpha$ struggles to break down the additional noise introduced during the regime transition within the 250-day rolling window.

Regime	Market Condition	Avg MSE (B)	Avg MSE (C)	Δ Improvement	$\Delta\alpha$ Effectiveness
2	Moderate Stability	0.000364	0.000356	+2.27%	Weak
3	Transitional	0.000348	0.000344	+1.22%	Weak
4	High Volatility (Crisis)	0.000784	0.000545	+30.50%	Strong
5	Recovery	0.000835	0.000880	−5.35%	Negative
6	High Uncertainty	0.000942	0.000877	+6.97%	Moderate

Table 5.3.1: Average Regime-wise MSE — Model B vs Model C (Random Forest)

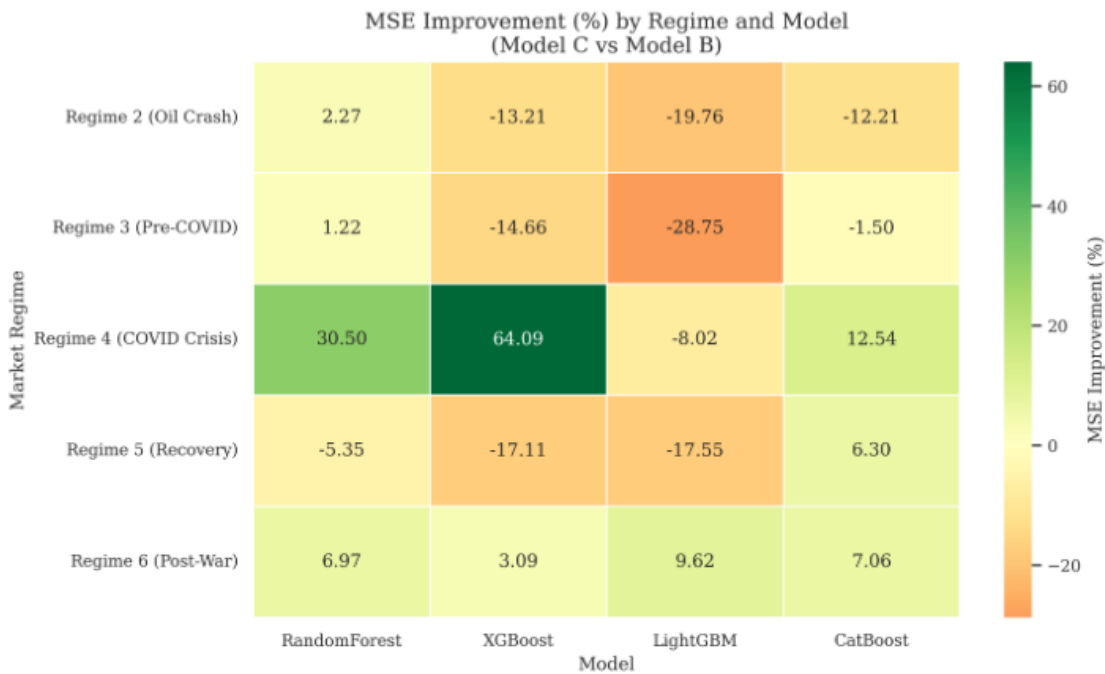


Fig 5.3 MSE Improvement % by Regime and Model

The figure 5.3 shows that the inclusion of multifractal features ($\Delta\alpha$) significantly improves model performance primarily during the high-volatility COVID crisis regime—most notably for XGBoost—while offering limited or negative gains in stable and recovery periods, confirming the regime-dependent effectiveness of multifractal information.

5.4 Volatility-Stratified Analysis

To complement the regime analysis, observations are split into high- and low-volatility subsets using the median 30-day rolling volatility as a threshold. Table 9 confirms the pattern that $\Delta\alpha$ is more effective under high volatility, regardless of the regime label.

Volatility Regime	Model	MSE	RMSE	MAE	MSE Improvement
Low Volatility	Model B	0.000291	0.01706	0.01340	—
Low Volatility	Model C	0.000274	0.01656	0.01298	+5.84%
High Volatility	Model B	0.001157	0.03401	0.02499	—

High Volatility	Model C	0.001085	0.03294	0.02388	+6.22%
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Table 5.4: Performance by Volatility Regime — Random Forest

The impact of $\Delta\alpha$ is marginal yet stable for both volatility environments when looking at the overall picture. However, when analyzing the performance of $\Delta\alpha$ within each volatility environment separately, the effect becomes more pronounced since Regime 4 is the dominant cause of the high-volatility performance enhancement. Thus, the findings suggest that the utility derived from multifractal properties depends on both the level and type of volatility.

5.5 Cross-Correlation Analysis

Pearson cross-correlation between Brent and BDTI returns is computed both globally and per regime to assess the linear co-movement structure that underpins the predictive relationship.

Regime	Description	Correlation (r)	Interpretation
1	Pre-Oil Crash	−0.114	Weak negative
2	Oil Crash	+0.003	Near zero
3	Pre-COVID	−0.011	Near zero
4	COVID Crisis	−0.196	Moderate negative
5	Recovery	+0.026	Near zero
6	Post-War	+0.053	Weak positive

Table 5.5: Pearson Correlation — Brent Return vs BDTI Return by Regime

The very small linear correlation seen in most regimes demonstrate that the predictive power of the BDTI on Brent prices lies mainly in its nonlinear nature. The negative correlation found in Regime 4 is in line with the concept of demand destruction when it comes to the coronavirus pandemic lockdown period; while oil demand dropped sharply, shipping rates fell as well.

5.6 Discussion

The findings clearly show that multifractal properties serve as conditional boosters with maximum predictive power in periods of high complexity and crisis. The following reasons support the results:

1. Scaling of complex processes during crisis events: Due to the COVID-19 shock, the Brent price series saw the concurrent effects of demand destruction, OPEC+ dispute-related supply disruption, and contagion in the financial markets, leading to non-homogeneous scaling in returns as seen through large $\Delta\alpha$ values, which contain additional discrimination power in comparison with volatility.

2. Noise prevalence in stable regime: In Regime 2 and 3 (oil price correction period), the market showed characteristics of a random walk. This implies that $\Delta\alpha$ was noisy in nature, and therefore, adding it as a feature added more noise to the model rather than enhancing prediction accuracy.

3. Model sensitivity: The superior prediction accuracy (+64.09%) recorded by XGBoost in Regime 4 when compared to Random Forest (+30.50%) and CatBoost (+12.54%) reveals that a gradient boosting algorithm with L1/L2 regularization is well-suited for extracting non-linear signals from $\Delta\alpha$ in highly unusual scenarios. LightGBM's poor predictive ability in Regime 4 can be attributed to the Gradient-based One-Side Sampling strategy.

CHAPTER 6

CONCLUSION AND FUTURE WORK

6. CONCLUSION AND FUTURE WORK

6.1 Conclusion

A multifractal and regime-based approach for predicting the Brent crude oil return series is proposed. The series consists of 3,250 daily data points from March 2013 to April 2026, encompassing six regimes based on multi-signal binary segmentation. Results include:

- Multifractal features ($\Delta\alpha$) show significant improvement during crises (regimes), yielding MSE reductions up to 64% (XGBoost, Regime 4 – COVID-19).
- During non-crisis and transitional periods, $\Delta\alpha$ does not offer any improvements, corroborating its conditional nature.
- CatBoost demonstrates the highest prediction performance (MSE = 0.000334), exceeding all baseline models, including ARIMA and GARCH.
- Importance assessment reveals $\Delta\alpha$ accounting for 21%–36% of the forecasted variance, behind lagged Brent (23%–52%) and BDTI (27%–40%).
- Analysis of variance (ANOVA) proves differences between regimes are significant ($F = 121.11$, $p \approx 0$), confirming regime identification.
- Cross-correlation results reveal minimal co-movement of linearly correlated Brent and BDTI series in five of six regimes, indicating intrinsic nonlinearity of BDTI features.

These findings confirm that financial predictability is inherently regime-dependent and that combining nonlinear complexity measures with regime-aware modelling represents a meaningful advance over standard time series approaches.

6.2 Future Work

Several extensions are proposed for future research:

- Deep learning implementation: LSTM networks and Transformer models equipped with attention could be extended to use rolling $\Delta\alpha$ as a feature that varies with time, possibly allowing for more extensive lags than the decision trees.
- Dynamic regime identification: Online change-point detection procedures (BOCPD) could be implemented to detect regimes dynamically and enable models to update their parameters accordingly in a timely manner.
- Feature expansion: Other shipping indices (BCTI, LNG rates), macroeconomic variables (VIX, DXY, EM spreads), and satellite-based tracking of vessels could enhance the pool of explanatory features.
- Asset class expansion: Testing the model on natural gas, coal, and energy stocks would reveal whether regime-based multifractal predictability is universal across assets.
- Ensemble learning: Ridge Regression meta-learner applied to the outputs of the four base learners might yield more reliable results due to complementarity between the models.

CHAPTER 7

REFERENCES

7. REFERENCES

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